Hannah Gonzalez Collison

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Baltimore, Maryland · USA Permanent Resident (does NOT need sponsorship)

RESEARCH INTERESTS

My research focuses on NLP, with interests in pretraining, data for LLMs, AI for science, and AI safety and alignment. My recent work uses LLM reasoning to assess publication impact, explores continual pretraining on scientific literature to improve long-context capabilities, and examines emergent misalignment in models.

EDUCATION

Johns Hopkins University — *Baltimore, MD*
Ph.D. Computer Science. Focused on NLP. — August 2025 - May 2029
Advisors: Daniel Khashabi and Benjamin Van Durme

University of Pennsylvania — *Philadelphia, PA*
M.S.E. Computer and Information Science. — August 2021 - December 2023
B.S.E. Computer and Information Science. Magna Cum Laude. — August 2019 - May 2023
Minors: Mathematics; Engineering Entrepreneurship.
Research Advisor: Chris Callison-Burch

RESEARCH EXPERIENCE

Johns Hopkins University — *Baltimore, MD*
Research Assistant, *with Daniel Khashabi and Benjamin Van Durme* — August 2025 - Present

University of Pennsylvania — *Philadelphia, PA*
Research Assistant, *with Eleni Miltsakaki and Chris Callison-Burch* — August 2021-December 2023

University of Pennsylvania — *Philadelphia, PA*
Research Assistant, *with Eleni Miltsakaki and Kostas Daniilidis* — May 2023-December 2023

Center For Neuroengineering and Therapeutics — *Philadelphia, PA*
Research Assistant, *with Kathryn Davis* — August 2021-May 2022

PROFESSIONAL EXPERIENCE

Microsoft Corporation, Azure AI Search — *Redmond, WA*
Software Engineer (Level 60) — January 2024 - August 2025

Microsoft Corporation, Core Search and AI Team — *Redmond, WA*
Software Engineer Intern — May 2023-August 2023

Microsoft Corporation, Core Search and AI Team — *Bellevue, WA*
Software Engineer Intern — May 2022-August 2022

Microsoft Corporation, Research and Development Engineering Team — *Redmond, WA*
Software Engineer + Product Manager Intern — May 2021-August 2021

SELECTED PUBLICATIONS

1. **Mid-training on citation-linked megapapers improves long-context capabilities.** In progress.
2. **Weird Generalization is Weirdly Brittle.** Miriam Wanner, **Hannah Collison**, William Jurayj, Benjamin Van Durme, Mark Dredze, William Walden. *BlackboxNLP@EMNLP 2026.*
3. 🏆 **Crystal: Characterizing Relative Impact of Scholarly Publications.** **Hannah Collison**, Benjamin Van Durme, Daniel Khashabi. *Under Review.*
4. 🏆 **Enhancing Human Summaries for Question-Answer Generation in Education.** **Hannah Gonzalez**, Liam Dugan, Eleni Miltsakaki, Zhiqi Cui, Jiaxuan Ren, Bryan Li, Shriyash Upadhyay, Etan Ginsberg, Chris Callison-Burch. *BEA at ACL 2023*, pages 108–118, Toronto, Canada.

5. **Automatically Generated Summaries of Video Lectures May Enhance Students' Learning Experience.** **Hannah Gonzalez**, Jiening Li, Helen Jin, Jiaxuan Ren, Hongyu Zhang, Ayotomiwa Akinyele, Adrian Wang, Eleni Miltsakaki, Ryan Baker, Chris Callison-Burch. *BEA at ACL 2023*, pages 382–393, Toronto, Canada.

6. **A Feasibility Study of Answer-Agnostic Question Generation for Education.** Liam Dugan, Eleni Miltsakaki, Shriyash Upadhyay, Etan Ginsberg, **Hannah Gonzalez**, Dayheon Choi, Chuning Yuan, Chris Callison-Burch. *Findings of ACL 2022*, pages 1919–1926, Dublin, Ireland.

7. **Deep Learning-Based Automated Segmentation of Resection Cavities on Postsurgical Epilepsy MRI.** Arnold Campbell, Ramya Muthukrishnan, Akash Pattnaik, Nishant Sinha, Adam Gibson, **Hannah Gonzalez**, Sandhitsu R. Das, Brian Litt, Dario J. Englot, Victoria L. Morgan, Kathryn A. Davis MD, Joel M. Stein. *NeuroImage: Clinical 36*, page 103154, 2022.

🏆 Best Poster Award

SELECTED GRANTS AND AWARDS

Walter Korn Award University of Pennsylvania, 2023
Awarded annually to one outstanding senior in the Moore School of Engineering at the University of Pennsylvania.

Dean's List University of Pennsylvania, 2022
Awarded annually to any student who achieves a combined GPA of 3.7 for the fall and spring semesters.

Vagelos Undergraduate Research Grant University of Pennsylvania, 2022
Grant that supports outstanding undergraduate research efforts.

Gutmann-Doyle Research Opportunities Fund Award University of Pennsylvania, 2022
Award to assist with activities associated with independent research projects.

CS Research Mentorship Program (CSRMP) Scholar Google Research, 2021
CSRMP matches students from historically marginalized groups with peers and a Google mentor to support their pursuit of computing research pathways.

SELECTED INVITED TALKS

RAGChat: Optimal retrieval with Azure AI Search (Presenter)	Microsoft Corporation, 2025
Women In Computer Science Big Tech Internship Panel (Panelist)	University of Pennsylvania, 2023
Stanford Research Conference (Poster Presenter)	Stanford University, 2023
Spring Research Symposium (Poster Presenter)	University of Pennsylvania, 2023
Grad Cohort for IDEALS Research Lightning Talk (Presenter)	Computer Research Association, 2023
CSRMP Alumni Panel Research Presentation (Presenter)	Google Research, 2023
OurCS (Poster Presenter)	Carnegie Mellon University, 2022
CSRMP Alumni Panel Research Presentation (Presenter)	Google Research, 2022
Penn Neuroscience Society Expo (Poster Presenter)	University of Pennsylvania, 2022
Finding the Right Fit Career Orientation (Panelist)	Education USA, 2022

TEACHING EXPERIENCE

University of Pennsylvania *Philadelphia, PA*
Teaching Assistant for Artificial Intelligence (CIS 5210) with Chris Callison-Burch August 2023-December 2023
University of Pennsylvania *Philadelphia, PA*
Course Staff for Engineering Entrepreneurship I (EAS 5450) August 2022-May 2023
University of Pennsylvania *Philadelphia, PA*
Instructor at Penn's Fife Academy August 2020-December 2020
University of Pennsylvania *Philadelphia, PA*
Teaching Assistant at Penn Engineering Summer Academy July 2020